

WHERE OUTRAM PARK STANDS

A status review by physics module

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SECTION 1

The Shape of the Thing

WHAT OUTRAM PARK IS

A multi-physics simulation workspace written in Rust, assembled mostly by porting established open-source engineering codes rather than inventing new ones.

- ◉ **38 crates, 1.27 million lines** of Rust.
- ◉ **1,676 commits**, first commit 25 November 2024.
- ◉ **8,453** declared test functions across the workspace.
- ◉ **11 of 38** crates are declared *mature*; the other 27 explicitly are not.

The prior throughout is “**this probably exists already**”. Most modules are ports of OpenFOAM (Weller et al. 1998), OpenMC (Romano et al. 2015), NJOY2016 (MacFarlane et al. 2017), CoolProp (Bell et al. 2014) and similar — with the upstream kept as the thing to be measured against.

WHAT “MATURE” IS ALLOWED TO MEAN

Maturity here is a *declared* status with a written bar, not a feeling about code quality. Each mature crate states its own acceptance bar and the class of evidence behind it:

Evidence class	What it means
Reference standard	Reproduces a published standard to stated tolerance
Cross-code	Agrees with the upstream code it was ported from
Experimental	Reproduces a measured experiment
Analytical / MMS	Matches closed-form solutions; convergence order verified
Unit + consistency	Limiting cases exact; published constants reproduced

A crate with no such written statement is **by definition not mature**. Only the maintainer may declare maturity.

READING THE TABLES THAT FOLLOW

- ◉ **kLOC** — thousands of lines of Rust in the crate, measured 2026-09-15.
- ◉ **Tests** — *declared* test functions counted statically (`#[test]` attributes). This is an inventory count, not a record of a passing run.
- ◉ **M** — ✓ if declared mature on 2026-09-13.

Measured on 2026-09-15: `cargo check --workspace --lib --tests` completes clean. The three most recently merged crates were run in full: **1,111 tests passed, 0 failed**.

Everything else below is inventory and declared status — stated as such, because the difference matters.

SECTION 2

Nuclear Data

NUCLEAR DATA: NJOY-OUTRAM-PARK-FORK

Crate	kLOC	Tests	M
njoy-outram-park-fork	119.9	824	✓

A pure-Rust port, in progress, of the NJOY2016 nuclear data processing system (MacFarlane et al. 2017).

Bar: agreement with NJOY2016 to **7 significant figures** — a cross-code bar, the strongest class available for a port.

Recently: the GROUPR and LEAPR modules were finished; THERMR's `calcem/sig1` paths were ported and its E' grid is now adaptively linearised; the LEAPR `isym` defect was fixed.

This is the deepest single module in the workspace and the one whose validation story is cleanest.

SECTION 3

Neutron Transport and Criticality

NEUTRON TRANSPORT: 97.8 KLOC, 603 DECLARED TESTS

Crate	kLOC	Tests	M
outram-mc-libs	74.7	392	✓
boon-lay	21.4	187	
outram-park-fork-onix	1.7	24	

outram-mc-libs ports the OpenMC Monte Carlo transport kernels (Romano et al. 2015). boon-lay carries Lagrangian transport and transmutation; outram-park-fork-onix is an early port of the ONIX depletion code (Troullioud de Lanversin, Kütt, and Glaser 2021) and is barely started.

Bar (outram-mc-libs): k_{eff} within **500 pcm** of the ICSBEP Godiva benchmark (*International Handbook of Evaluated Criticality Safety Benchmark Experiments* 2021).

A separate gate covers the **HTR-10 double-heterogeneity difference** (*Evaluation of High Temperature Gas Cooled Reactor Performance: Benchmark Analysis Related to Initial Testing of the HTTR and HTR-10* 2003) — that difference alone, deliberately, and nothing else.

NEUTRON TRANSPORT: WHAT THE LAST MONTH ACTUALLY DID

The interesting work here was *finding errors*, not adding features:

- ◉ Surface tracking computed nested-universe surface normals about the **wrong centre**. Fixing it moved LEU-COMP-THERM-008 from +2341 pcm to +**37 pcm** against a measured critical, and moved the hex lattice by 1769 pcm.
- ◉ The delta-tracking majorant **under-bounded** Σ_t below its grid floor by up to $9\times$ — an outright correctness bug in the sampling.
- ◉ The ring-RPT pebble case now reproduces exact tracking to +37 pcm at $9\times$ statistics.
- ◉ A previously claimed throughput speed-up was **measured and refuted** (GH #205); reproducibility was confirmed instead.

The 500 pcm bar is set at *what the code achieves today*, not at what criticality safety work should eventually demand.

SECTION 4

Reactor Kinetics and Coupled Systems

KINETICS AND COUPLING: 30.6 KLOC, 208 DECLARED TESTS

Crate	kLOC	Tests	M
teh-o-prke	12.5	28	✓
bedok	14.1	155	
outram-park-fork-moltres	3.0	20	
nee_soon	1.0	5	

teh-o-prke solves the point reactor kinetics equations and drives the teaching simulators. bedok is systems-level coupling: 3-D nodal diffusion against thermal hydraulics. outram-park-fork-moltres follows Moltres ([Lindsay et al. 2018](#)) for molten-salt neutronics on the outram-foam finite-volume layer.

Bar (teh-o-prke): published β reproduced; PRKE limiting cases exact.

Honest note: this is the *thinnest* of the eleven mature crates. It has **no analytical transient validation** — its own file says so, and says what would fix it.

SECTION 5

Thermal Hydraulics and CFD

THERMAL HYDRAULICS AND CFD: 344.9 KLOC, 1,630 DECLARED TESTS

Crate	kLOC	Tests	M
tuas_boussinesq_solver	167.9	381	✓
outram-foam-basic-lib	63.1	586	✓
outram-foam-appbuilder-lib	46.9	323	✓
tampines	32.8	113	
outram-foam-mesh	13.1	72	
outram-park-fork-cfmesh	9.6	67	
outram-foam-multiphase	7.1	67	
outram-foam-turbulence-lib	3.1	16	
outram-foam-cli	1.3	5	

The largest module by volume, and the only one with three independently mature crates.

THERMAL HYDRAULICS: THE THREE BARS

- ⊙ `tuas_boussinesq_solver` — CIET facility outlet temperature within **0.2 °C**; Gnielinski correlation within **2%** (Gnielinski 1976; Zweibaum 2015). This is the workspace's only *experimental* validation leg.
- ⊙ `outram-foam-basic-lib` — conservation to 10^{-12} ; convergence order matches theory.
- ⊙ `outram-foam-appbuilder-lib` — Sod shock tube (Sod 1978) against the exact Riemann solution, L_2 within 5% of peak; a timed series was added recently.

Meshing (`outram-foam-mesh`, and the `cfMesh` port (Juretić 2015)), turbulence and multiphase are ported but **not declared** — turbulence in particular is 3.1 kLOC against OpenFOAM's full RAS/LES/DES catalogue.

SECTION 6

Thermophysical Properties and Chemistry

PROPERTIES AND CHEMISTRY: 330.6 KLOC, 2,187 DECLARED TESTS

Crate	kLOC	Tests	M
tampines-steam-tables	166.8	1097	✓
outram-park-fork-dwsim-libs	92.7	721	✓
outram-park-fork-coolprop	69.4	363	
outram-park-fork-thermochimica	1.7	6	

The most heavily tested module in the workspace — appropriately, since everything else calls into it.

Bar (steam tables): IF97 region equations to 10^{-8} against published IAPWS values (Wagner et al. 2000); flash tables held to a looser bar (0.5% volume, 8% thermal conductivity).

PROPERTIES: RECENT WORK, AND ONE CAVEAT THAT MATTERS

Steam tables. The (ρ, h) flash now works by inverting the IF97 backward equations; (p, h) , (p, s) and (h, s) cover Region 5. Two *negative* results were recorded on where (ρ, h) does not belong, with conditioning measured on the answer rather than on a local slope.

DWSIM port — read this before citing it. The bar is agreement with upstream DWSIM (DWSIM Project 2026) to 4 significant figures. Upstream has since been built and run headless, and the Peng–Robinson Z agrees to **6 significant figures**.

But that is **the EOS layer only**. The flash-layer comparison the bar also names is still open, and **0 of the 107 rows** in its port-coverage matrix are marked PORTED + VALIDATED.

outtram-park-fork-thermochimica (Piro et al. 2013) is a stub: 1.7 kLOC, 6 tests.

SECTION 7

Fuel Performance and Solid Mechanics

FUEL AND STRUCTURES: 77.4 KLOC, 649 DECLARED TESTS

Crate	kLOC	Tests	M
outram-park-fork-offbeat	61.3	550	
farrer-park	16.1	99	✓

outram-park-fork-offbeat ports the OFFBEAT fuel behaviour solver (Scolaro et al. 2020) — substantial at 61 kLOC, but **not declared mature**. farrer-park is the finite-element layer underneath it.

Bar (farrer-park): MMS L_2 convergence order within 0.15 of theory per element; patch test to 10^{-12} ; Lamé *displacement* within 1%, *stress* on observed order.

This crate is the clearest illustration of what the maturity label does and does not promise.

- ⦿ It carries a **known uncured defect**: shear locking, measured at **66.7% too stiff** on Quad4 elements at element aspect ratio 4.
- ⦿ Its maturity therefore says *the numerics it implements converge at the theoretical rate* — **not** that every element is fit for bending.
- ⦿ It is the only roster entry declared **within days of the crate first existing**, on an analytical/MMS bar with **no cross-code and no benchmark leg at all**.

Declared mature 2026-09-13 by maintainer decision, with all of the above written into the declaration rather than omitted from it.

SECTION 8

Granular, Subsurface and Process

Crate	kLOC	Tests	M
outram-park-fork-pflotran	27.6	278	
outram-park-fork-liggghts	9.6	72	✓

Granular DEM (Kloss et al. 2012) — pebble-bed contact mechanics and thermal DEM. Its bar covers the **integrator and contact laws against closed-form solutions**. The declaration states plainly that **granular physics is NOT validated**: the time integration is right, the emergent bed behaviour is unproven.

Subsurface (Hammond, Lichtner, and Mills 2014) — flow and reactive transport, 27.6 kLOC and 278 tests, not declared. Sound in size, untested against the upstream.

PROCESS CONTROL AND DIGITAL TWIN: 155.8 KLOC

Crate	kLOC	Tests	M
outram-park-digital-twin-engine	101.7	577	
outram-blender	40.1	474	
chem-eng-real-time-...-simulator	14.0	32	✓

The twin engine drives the visual process objects behind the HTGR and iPWR simulators; outram-blender ([Blender Foundation 2026](#)) is the headless mesh-authoring frontend. Both are large and well-tested but **not declared**.

Bar (process control): discretisations exact at sample points against closed-form transfer functions.

Honest note: the Scilab half of that crate's evidence lives in the maintainer's dissertation, *not* in this repository. Its recorded bar is written against the analytical tests that *are* reproducible here.

SECTION 9

Supporting Numerics

SUPPORTING NUMERICS AND TOOLING: 71.0 KLOC, 919 TESTS

Crate	kLOC	Tests
kovan family (7 crates)	57.0	810
petir	6.0	42
raffles	5.7	34
outram-park-mpi	2.3	33

- ◉ `petir` — `no_std` polynomials, transforms, integration and roots. Its fast `exp/ln/powf` are now the default route for the steam tables.
- ◉ `raffles` — risk analysis and ensemble simulation.
- ◉ `outram-park-mpi` — a pure-Rust MPI-3 API subset (*MPI: A Message-Passing Interface Standard, Version 3.1 2015*).
- ◉ `kovan` — the knowledge layer: discovery, literature, semantics, codegen, metrics.

None of these are physics modules, and none are declared mature.

SECTION 10

Where It Stands

THE HONEST SUMMARY

Module	kLOC	Tests	Mature
Thermal hydraulics & CFD	344.9	1630	3 of 9
Properties & chemistry	330.6	2187	2 of 4
Process control & twin	155.8	1083	1 of 3
Nuclear data	119.9	824	1 of 1
Neutron transport	97.8	603	1 of 3
Fuel & solid mechanics	77.4	649	1 of 2
Supporting numerics	71.0	919	0 of 10
Kinetics & coupling	30.6	208	1 of 4
Subsurface	27.6	278	0 of 1
Granular DEM	9.6	72	1 of 1
Total	1265.2	8453	11 of 38

WHAT IS EXPLICITLY NOT VALIDATED

Stated plainly, because the roster states it plainly:

- ◉ **DWSIM port** — EOS layer only. 0 of 107 port-coverage rows are PORTED + VALIDATED. Flash layer open.
- ◉ **Granular DEM** — integrator and contact laws validated; **granular physics is not.**
- ◉ **farrer-park** — shear locking uncured, 66.7% too stiff on Quad4 at aspect ratio 4; no benchmark leg.
- ◉ **teh-o-prke** — no analytical transient validation.
- ◉ **outram-mc-libs** — 500 pcm is today's achievement, not a criticality-safety standard.
- ◉ **27 of 38 crates** carry no maturity declaration at all, including 61 kLOC of fuel performance and 69 kLOC of CoolProp.

WHERE THE WORK IS GOING

- ◉ **Close the DWSIM flash layer.** The EOS leg is done to 6 significant figures; the 107-row matrix is the remaining work, and it is the clearest single gap on the roster.
- ◉ **Give teh-o-prke an analytical transient leg,** and farrer-park a cross-code or benchmark leg.
- ◉ **Cure shear locking** — or document the element as unfit for bending and select around it.
- ◉ **Tighten the Monte Carlo bar** below 500 pcm now that the nested-universe and majorant defects are fixed.
- ◉ **Declare or retire** the large undeclared ports: OFFBEAT, CoolProp, PFLOTRAN, the twin engine.

The discipline that produced the last month's results — measuring claims and *retracting* the ones that fail — is the part worth keeping.

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